House Price Predictor

Linear Regression Model — Project Report

AI & Machine Learning Internship — Task 1 | Maincrafts Technology

Dataset: California Housing | Algorithm: Linear Regression | Language: Python 3.11

1. Introduction

This report presents a complete machine learning project that builds and evaluates a Linear Regression model to predict median house prices across California neighborhoods. The project follows the standard ML lifecycle: data loading, exploratory data analysis (EDA), preprocessing, model training, evaluation, and reporting.

Objective

Develop a reproducible ML pipeline using Python and scikit-learn that accurately predicts the median house value for a given California block group, based on eight socioeconomic and geographic features.

Dataset Overview

The California Housing dataset is a well-known benchmark dataset derived from the 1990 U.S. Census. It is built directly into scikit-learn and contains 20,640 records with 8 input features and 1 continuous target variable.

Feature	Description	Type
MedInc	Median income in block group	Numerical
HouseAge	Median house age in block group	Numerical
AveRooms	Average number of rooms per household	Numerical
AveBedrms	Average number of bedrooms per household	Numerical
Population	Block group population	Numerical
AveOccup	Average household occupancy	Numerical
Latitude	Block group latitude	Geographical
Longitude	Block group longitude	Geographical
MedHouseVal	**Median house value (TARGET)**	**Target**

Table 1: Dataset features and descriptions

2. Exploratory Data Analysis (EDA)

Data Quality Check

Before building any model, the dataset was thoroughly examined for data quality issues. The following checks were performed:

✓ **Missing Values:** None found across all 9 columns. The dataset is complete and requires no imputation.

✓ **Data Types:** All 8 features are continuous numerical values. No categorical encoding needed.

✓ **Dataset Shape:** 20,640 rows and 9 columns (8 features + 1 target variable).

✓ **Duplicates:** No duplicate records detected in the dataset.

Key EDA Findings

• House prices are right-skewed, with most values between $100k and $300k. A hard cap at $500k is visible in the data, suggesting price truncation in the original census.

• Median Income (MedInc) shows the strongest positive correlation with house price at 0.69, making it the most predictive feature.

• Latitude and Longitude show moderate negative correlations (-0.14 and -0.05), reflecting geographic price variation across California.

• AveRooms and AveBedrms show weak positive correlations (0.15 and 0.05), suggesting room count alone is not a strong price driver.

• Population and AveOccup show near-zero correlations with price, indicating block density has minimal direct effect.

• Income distribution is right-skewed, with most households earning between $20k and $60k annually.

Feature Correlation with Target (MedHouseVal)

Feature	Correlation	Strength
MedInc	**0.688**	**Strong Positive**
AveRooms	0.151	Weak Positive
HouseAge	0.106	Weak Positive
AveBedrms	0.048	Very Weak
Population	-0.025	Negligible
AveOccup	-0.024	Negligible
Latitude	-0.145	Weak Negative
Longitude	-0.046	Very Weak Negative

Table 2: Pearson correlation of each feature with target variable

3. Model Development

Preprocessing

- All 8 features were selected as inputs — no features were dropped as all showed some relationship with the target.
- Data was split 80/20: 16,512 samples for training and 4,128 samples for testing (random_state=42 for reproducibility).
- StandardScaler was applied to normalize features to zero mean and unit variance, preventing high-magnitude features from dominating the model.
- The scaler was fit only on training data and then applied to both train and test sets to prevent data leakage.

Algorithm: Linear Regression

Linear Regression models the relationship between features and the target as a weighted linear combination. The model learns one coefficient (weight) per feature during training by minimizing the sum of squared residuals (Ordinary Least Squares).

Learned Feature Coefficients

Feature	Coefficient	Interpretation
MedInc	+0.8296	Higher income strongly increases price
Latitude	-0.8998	Further north tends to decrease price
Longitude	-0.8707	Further east tends to decrease price
AveBedrms	+0.3057	More bedrooms slightly increases price
AveRooms	-0.2654	More rooms slightly decreases price
HouseAge	+0.1188	Older houses slightly more expensive
AveOccup	-0.0392	Higher occupancy slightly lowers price
Population	-0.0045	Negligible effect on price

Table 3: Model coefficients after training on scaled features

4. Evaluation Results

Performance Metrics

Metric	Value	Meaning
MAE (Mean Absolute Error)	0.5332	Average prediction error of $53,320

RMSE (Root Mean Sq. Error)	0.7456	Typical prediction error of $74,560
R<super>2</super> (R-Squared Score)	0.5758	Model explains 57.6% of price variation

Table 4: Model evaluation metrics on the test set (4,128 samples)

Interpretation

- **MAE = $53,320:** On average, the model's house price predictions are off by about $53k. For California house prices ranging from $15k to $500k, this is a reasonable baseline error.
- **RMSE = $74,560:** The RMSE is higher than MAE, meaning there are some larger errors being penalized. This is typical when a dataset has capped values (at $500k) that the model struggles with.
- **R-Squared = 0.5758:** The model explains 57.6% of the variance in house prices. The remaining 42.4% is driven by factors not captured in the dataset such as school quality, crime rates, and neighborhood amenities.

Plot Descriptions

- **Actual vs Predicted Scatter Plot:** Points cluster around the ideal red diagonal line, confirming the model captures the general price trend. Scatter increases at higher price ranges, indicating the model is less confident for expensive houses.
- **Residual Distribution:** The residuals form a near bell-shaped distribution centered close to zero, confirming the model has no systematic bias. The slight right skew is caused by the $500k price cap in the dataset.
- **Residuals vs Predicted:** Residuals are mostly scattered randomly around the zero line. Some patterns appear at higher predicted values, suggesting non-linear relationships that Linear Regression cannot fully capture.

5. Improvement Ideas

While Linear Regression provides a solid baseline, several improvements can significantly boost performance:

- **Ridge / Lasso Regression:** Regularized linear models that penalize large coefficients, reducing overfitting. Ridge uses L2 penalty; Lasso uses L1 and can perform feature selection automatically.
- **Polynomial Features:** Adding squared or interaction terms (e.g. MedInc x AveRooms) can help the model capture non-linear relationships that plain linear regression misses.
- **Random Forest Regressor:** An ensemble of decision trees that can model complex non-linear relationships. Typically achieves R-squared of 0.80+ on this dataset, a significant improvement.
- **XGBoost / Gradient Boosting:** State-of-the-art boosting algorithms that iteratively correct prediction errors. Likely to achieve R-squared of 0.83+ on this dataset.
- **Feature Engineering:** Creating new features such as rooms-per-person (AveRooms / AveOccup) or income-per-room (MedInc / AveRooms) can improve predictive power.

- **Outlier Removal:** Removing records where house prices are capped at $500k (about 965 records) could improve model accuracy on normal price ranges.
- **Cross-Validation:** Using k-fold cross-validation instead of a single train/test split provides a more reliable estimate of model performance and reduces variance.
- **Hyperparameter Tuning:** Using GridSearchCV or RandomizedSearchCV to find optimal model parameters for regularized or ensemble models.

6. Conclusion

This project successfully demonstrated the complete machine learning workflow from raw data to a trained and evaluated predictive model. A Linear Regression model was built on the California Housing dataset achieving an R-squared of 0.576, MAE of $53.3k, and RMSE of $74.6k on unseen test data.

The most important finding from EDA was that Median Income is by far the strongest predictor of house prices with a correlation of 0.69. Geographic features (Latitude, Longitude) also contributed significantly to the model via their learned coefficients.

The model serves as a strong baseline for further experimentation. Future iterations using Random Forest or XGBoost are expected to significantly improve upon these metrics. The trained model and scaler have been saved as pickle files for future inference on new data.

FINAL PROJECT SUMMARY
Dataset: California Housing | 20,640 records | 8 features | No missing values
Model: Linear Regression | Train: 16,512 | Test: 4,128 | Scaler: StandardScaler
Results: MAE $53.3k | RMSE $74.6k | R-Squared 0.576 (57.6% variance explained)
Top Feature: MedInc (correlation 0.69) | Saved: .pkl model + scaler